BGIN Meeting Report

Date: TBD
Location: Virtual Meeting

Executive Summary

The meeting focused on the critical issues surrounding the development of a standardized taxonomy for the crypto industry, including regulatory frameworks, information sharing mechanisms, and the role of analytics and digital forensics. Participants discussed the need for clear definitions and consistent regulatory guidelines to combat illicit activities while fostering innovation. Key recommendations included developing a standardized taxonomy, establishing robust information sharing protocols, providing guidance on the use of analytics and digital forensics, and creating clear and consistent regulatory frameworks.

Key Discussion Points

1. **Taxonomy and Definitions**:
   - The need for standardized definitions and taxonomy in the crypto space was emphasized to facilitate information sharing, regulation, and compliance.
   - Participants discussed the importance of aligning these definitions with existing regulatory frameworks and international standards (e.g., ISO).
2. **Information Sharing**:
   - The meeting highlighted the importance of establishing mechanisms for information sharing between regulators, industry players, and other stakeholders to combat illicit activities effectively.
   - Concerns about data protection and confidentiality were raised, particularly in cross-border contexts.
3. **Analytics and Digital Forensics**:
   - There was a strong emphasis on the role of analytics and digital forensics in detecting and preventing illicit activities in the crypto space.
   - Participants discussed the need for guidance on the use of these tools to ensure they are reliable and admissible in legal contexts.
4. **Regulatory Frameworks**:
   - The meeting underscored the need for clear and consistent regulatory frameworks that balance innovation with risk management.

- Global consistency was identified as crucial for facilitating cross-border transactions and information sharing.

Action Items and Next Steps

1. **Develop a Standardized Taxonomy**:
   - Form a working group to develop a standardized taxonomy for crypto-related terms.
   - Conduct a gap analysis of existing regulatory frameworks and taxonomy to identify areas that require improvement.
2. **Establish Information Sharing Mechanisms**:
   - Develop mechanisms for information sharing between regulators, industry players, and other stakeholders.
   - Address concerns about data protection and confidentiality through clear guidelines and agreements.
3. **Provide Guidance on Analytics and Digital Forensics**:
   - Create guidance documents on the use of analytics and digital forensics to detect and prevent illicit activities in the crypto space.
   - Engage with legal experts to ensure that these tools meet regulatory standards for evidence admissibility.
4. **Develop Clear Regulatory Frameworks**:
   - Work towards developing clear and consistent regulatory frameworks that balance innovation with risk management.
   - Collaborate with international organizations to promote global consistency in regulatory approaches.

Detailed Session Summary

- **Introduction**:
  - The meeting began with an overview of the challenges in creating a comprehensive taxonomy for blockchain analytics and digital forensics.
  - Participants discussed the importance of aligning definitions across different jurisdictions and regulatory frameworks.
- **Taxonomy and Definitions**:
  - It was noted that while there is no lack of definitions, there is a significant amount of conflicting information in the crypto industry.
  - A mapping exercise was proposed to understand what definitions are already available and identify gaps where new definitions are needed.
  - The need for global consistency in regulatory frameworks and taxonomy was emphasized to facilitate cross-border transactions and information sharing.
- **Information Sharing**:

  - Participants discussed the importance of establishing mechanisms for information sharing between regulators, industry players, and other stakeholders.
  - Concerns about data protection and confidentiality were raised, particularly in cross-border contexts.
  - It was suggested that clear guidelines and agreements are necessary to address these concerns.
- **Analytics and Digital Forensics**:
  - The role of analytics and digital forensics in detecting and preventing illicit activities in the crypto space was highlighted.
  - Participants discussed the need for guidance on the use of these tools to ensure they are reliable and admissible in legal contexts.
  - The importance of developing or acquiring these capabilities among industry players was also emphasized.
- **Regulatory Frameworks**:
  - The meeting underscored the need for clear and consistent regulatory frameworks that balance innovation with risk management.
  - Global consistency was identified as crucial for facilitating cross-border transactions and information sharing.
  - Participants discussed the importance of international cooperation between regulators and industry players to combat illicit activities that often have a global reach.
- **Next Steps**:
  - A working group will be formed to focus on developing a taxonomy that aligns with regulatory requirements and international standards.
  - The mapping exercise will begin immediately, and a machine-readable format for describing vulnerabilities will be developed.
  - Collaboration with ISO TC307 will be initiated to push for updates in existing standards.
- **Closing Remarks**:
  - The meeting concluded with an invitation to the network reception at 5:30 PM, which is a five-minute walk from the current location.
  - Participants were reminded to depart by 5:20 PM and take 10 minutes to arrive at the venue.

Appendix: Full Meeting Transcript (Anonymized)

```
16.5 --> 20.3:  Hello everyone, we're going to get started with this sess
ion now.
21.1 --> 27.3:  This session is on a common lexicon for threats and harms
```

in the crypto space.
28.6 --> 50.6: The first time we did this session was at block 13, which was actually an extension on some previous work that we were doing around interactions with the blockchain analytics companies to try and build this sort of like common language of information sharing between them for the analytics.
51.3 --> 58.4: And then that conversation progressed into, but what's the difference between analytics and forensics?
58.9 --> 62.7: And we did some work on that for about a year.
63.7 --> 75.1: And then that informed this more specific outcome, which is finding this common lexicon, which hopefully solves both of those problems.
75.5 --> 81.8: One being having this sort of common language and potentially database for people to share.
82.5 --> 86.2: And secondly, getting the terminology right between forensics and analysis.
86.9 --> 107.2: So all of the work of the IKP working group and now the FASE working group inclusive as well around the threats and harms and the identification of what is an analytic understanding of blockchain information and what can be used in forensics and in court.
107.2 --> 117.3: And then, of course, we get into the sort of the other side, which is not around the AML CTF, but around the market integrity.
117.7 --> 124.0: And fortunately, we have Han here today to speak about that and provide some insight.
129.7 --> 144.4: So I think we will get started just with that presentation to set the set the tone and the theme around the sort of market based threats and harms and mitigations that are in place.
144.4 --> 158.8: And then hopefully that will give us some some grease to transition the conversation into the practical outcome of coming up with this common lexicon that if we can do will be a great benefit to the industry.
161.5 --> 162.9: How long do I have?
167.4 --> 172.9: No, I definitely I'll try to keep it at around 10 minutes just to get us kicked off.
172.9 --> 174.7: And all right, great.
174.8 --> 176.5: So I'm just going to share my screen here.
179.6 --> 181.0: Great to be here again.
181.1 --> 187.9: I think I was in Tokyo for BGN 11, Block 11 two years ago and recently in Georgetown.
188.6 --> 191.4: It's just always inspiring to see the work you all are doing.
192.5 --> 193.6: Sorry. There we go.
193.9 --> 194.8: Are you seeing my screen?
195.3 --> 197.3: No. Try again.
200.1 --> 201.0: Sorry for that.
202.9 --> 212.8: Anyway, I'll start by saying that Solidus Labs, the firm I am one of the founders of, is a crypto native trade surveillance solu

tion.
213.0 --> 227.5: So we've been very focused for the past almost a decade
 on, you know, typologizing, understanding, defining and redefining marke
t manipulation, market abuse and increasingly other forms of financial cr
ime.
228.1 --> 232.8: We kind of see a lot of them, you know, becoming less a
nd less separate.
233.1 --> 236.5: So oftentimes manipulation, fraud are more mixed in dig
ital asset, et cetera.
237.5 --> 239.4: There we go. I think now it's working.
241.2 --> 247.9: So, yeah, you know, I'm going to skip like a lot of the
 kind of company presentation and stuff.
248.0 --> 250.0: But we've been around at this point for about half a de
cade.
250.0 --> 254.6: We work very closely with a lot of regulators and some
of the largest platforms in the world.
255.3 --> 267.6: We did extend to start applying some of our approaches
to detecting market manipulation in the digital asset space into securiti
es and derivatives recently.
267.7 --> 278.6: So you're really kind of starting to see how there's a
lot that Wall Street can learn from digital assets, not only, you know, o
n tokenization and, you know, bring those efficiencies, but also in fight
ing crime.
279.0 --> 289.5: One thing I'm not going to really delve deeply into the
 slide, I just find that proportion is really helpful sometimes when thin
king about financial crime on chain in the digital asset space.
290.5 --> 294.1: It's not like these issues are resolved in traditional
finance.
294.8 --> 301.1: Now, I'm going to give a lot of data and numbers and ex
amples of forms of market abuse that happen in digital assets.
301.2 --> 304.1: Some of them are quite scary and there are reasons for
it.
304.3 --> 307.9: We're all here in order to better define those issues a
nd how to deal with them.
307.9 --> 316.0: But, you know, some of the numbers up there, I won't go
 through them just to remind everyone that it's not like insider trading.
316.1 --> 317.1: It's not like money laundering.
317.3 --> 326.7: It's not like fraud doesn't happen and happen very ofte
n in traditional finance, despite immense investment and not just by gove
rnments, but also by industry through compliance.
328.1 --> 335.6: All right. One thing I just want to emphasize, it's ver
y fundamental to the way we approach a space and it's also very relevant
for specifically Japan.
337.0 --> 340.9: Because, you know, I like to reference the study by the
 FSA from 2023.
341.5 --> 345.2: You know, we like to think we like to emphasize on chai
n activity.

345.7 --> 347.2: It's transparency. It's immutability.
347.3 --> 354.7: All of these advantages is huge advantages for, you kno
w, fighting crime, doing compliance more efficiently.
355.3 --> 366.4: But we sometimes forget that while the assets are decen
tralized, they're issued on a blockchain in a permissionless environment,
 around 80 percent of the activity is not happening on chain.
367.0 --> 381.0: So, you know, if all of this forms of crime and concern
s and illicit activity was happening on chain, it would be a specific cha
llenge to understand it and detect it and typologize it, etc.
381.6 --> 385.5: If all of it was off chain, it would still be a little
different than traditional finance, but more similar.
385.6 --> 394.8: The fact that it's cross on and off chain really create
s some unprecedented challenges, especially in cross market and cross on
and off chain.
394.8 --> 412.1: I mentioned the FSA was very early in pointing out that
 in its study in July 2023, where essentially it asked the question, is o
n chain data enough to understand the health, integrity, security of dece
ntralized assets and their ecosystems?
412.3 --> 414.9: And the bottom line answer was, no, you need to constan
tly look beyond.
415.5 --> 420.7: I love this chart because as long as I've been staring
at it, it's still hard for me to really understand or explain it.
420.7 --> 422.1: But it's a very complex problem.
423.2 --> 432.9: You know, again, I'm not going to go too much into it,
but our approach is designed to address that, collecting both on chain an
d off chain data of various kinds, not just trading data, to be clear.
433.0 --> 434.9: That was also something that was very clear in the FSA
study.
435.6 --> 439.4: And that's how we use our technology to detect stuff.
439.9 --> 445.8: A lot of the stuff I'm going to share has been publishe
d, covered by media, based on our reports.
446.8 --> 450.7: I'm going to dive into a few examples here, but just to
 say that all of this is available.
450.7 --> 453.5: I'll also mention this is I have like 65 slides here.
453.5 --> 454.7: I'm not going to go through all of them.
454.9 --> 456.2: Very happy to share it.
456.2 --> 460.6: And we also have a lot of it available in Japanese, if
anyone is interested.
463.0 --> 466.1: So I'll start with insider trading.
466.9 --> 470.8: It's, you know, perhaps one of the most well-known form
s of market abuse.
471.1 --> 472.8: Definitely top of mind for regulators.
473.0 --> 480.6: I like to start with it because I don't think there's a
 regulator in the world or really anyone involved in trading that doesn't
 know that it's wrong and that it's a problem.
480.7 --> 490.8: I will say before I kind of address the big elephant of
 that number, which is very high on the slide, that, again, it still happ

ens in traditional finance quite often.
490.9 --> 510.7: There's a study by the University of Technology Sydney from a few years ago that analyzed massive data from U.S. securities markets and found that insider trading still happens leading to 20% of quarterly earning reports and 5%, sorry, 20% of M&A.
510.7 --> 514.7: activity and 5% of a quarterly earning report.
514.8 --> 523.7: So that being said, this unique market structure, assets that are impacted by both on-chain and off-chain activity, creates a challenge.
523.9 --> 525.7: This is a study we published a couple of years ago.
525.8 --> 529.5: We looked at three of the largest crypto platforms.
529.8 --> 531.4: The study was aggregated.
531.5 --> 532.5: It wasn't about the platforms.
532.7 --> 533.6: It was about the issue.
534.3 --> 546.8: Essentially, we looked at instances where DeFi available tokens or assets, essentially any digital assets before it gets listed on a centralized exchange, get listed on those large platforms.
548.0 --> 550.9: 234 listings of ERC-20 tokens.
551.7 --> 560.2: And we found that in 56% of cases, there was very clear evidence of a potential insider trading on the on-chain activity.
560.7 --> 571.4: Now, the methodology here, which we developed together with one of our regulator clients in the U.S., really focuses on looking at trading activity on-chain.
571.6 --> 573.0: That, of course, is transparent.
573.4 --> 583.8: You can look at basically any DEX activity you'd like and coalescing it with, in this case, off-chain activity, the listing itself or news about the listing.
584.3 --> 586.7: So essentially, this is just a very classic example.
586.7 --> 593.6: You can see this is a TVK listing on Binance.
593.9 --> 611.0: You can very easily see consistent buys at a certain, at a more or less stable price, leading to a news event, an announcement that is going to be listed, price increase immediately and very, very convenient sales.
611.2 --> 614.7: So that is classic insider activity.
614.7 --> 618.0: By the way, it's important to say, you know, we're not the law.
618.3 --> 622.9: Ultimately, that's where regulators come in, investigate market manipulation.
623.5 --> 625.1: Unlike AML, it's a bit less binary.
625.5 --> 626.9: There's a lot of intent element there.
627.4 --> 637.3: But obviously, by looking at all this data, coalescing the right data points across on-chain and off-chain activity, you can get a very good sense of where to look.
637.9 --> 639.7: Again, there's a lot of other information here.
640.5 --> 645.4: For example, one thing that was interesting is that this is the trading activity itself.
645.5 --> 656.0: The problematic trading activity is purely on-chain, bu

t a fifth of the wallets that we identified as involved in it were connec
ted to centralized exchanges as well.
656.1 --> 663.1: So also just very important, you cannot really separate
 the on-chain and the off-chain activity, the regulated and then less reg
ulated activity.
663.8 --> 665.0: And again, I'm happy to share this.
665.1 --> 667.2: There's a lot of more information here.
669.7 --> 675.7: You know, the pump and dump and cross-venue pump deviat
ion is something that we're seeing more and more.
675.8 --> 678.6: And it's oftentimes a big problem for centralized platf
orms.
679.7 --> 695.4: You know, this example refers to an AMP-USD trading pai
r where on the centralized platform, a trader, you know, was able to pump
 the token very aggressively for 15 minutes.
695.9 --> 701.1: And then increased the price by 20 percent and dumped i
t very quickly.
701.3 --> 717.2: What's interesting here is, again, you're kind of seein
g some of the, you know, one of the big issues with this cross-on and off
-chain reality with the fact that most of the trading activity and the pr
ice formation happened in fragmented, centralized environments.
717.2 --> 729.2: Is that, you know, at that point, you know, unless you'
re really looking very closely at the pricing, not only at your own excha
nge, but also in the market, reference exchanges, etc.
729.3 --> 734.6: And there's a lot of work that needs to be done in term
s of kind of like deciding what's a reference exchange, what it should be,
 etc.
735.5 --> 744.6: Then you suddenly have a situation where a specific ass
et deviates in price very significantly from, you know, the broader marke
t.
746.4 --> 774.0: You know, just another example, a touting, so social me
dia or other kind of web activity related pumping, you know, we realized
very early on that in order to be able to effectively address some of the
se concerns, you have to include another kind of off-chain data that peop
le really don't think about it as an off-chain piece of data, but news, r
ight, lives on Web 2.0.
774.6 --> 777.4: For now, at least, depends who you ask.
777.8 --> 778.7: It probably will forever.
779.3 --> 795.4: But, you know, sometimes you just are not able to expla
in these very dramatic price changes like this instance, unless you're al
so looking at a lot of the Internet and where there might be activity tha
t could impact the price.
795.4 --> 809.0: So in this particular case, you know, a trader was trad
ing Uni, right, UniSwaps token on a major centralized exchange, obviously
 had some sort of a connection, or at least it seemed obvious.
809.1 --> 810.6: Obviously, all of this needs to be investigated.
810.9 --> 816.0: And we often, you know, we support our clients with inv
estigations, but obviously they have their own teams.

816.3 --> 817.9: So they went and investigated it.
819.0 --> 826.5: And by the way, of course, like many, many other things,
 you know, LLMs, the ability to create content really quickly makes this
even more challenging.
826.6 --> 838.2: But the bottom line is that it would have been very eas
y not to be able to explain this or at least have even enough evidence to
 launch an investigation unless you were also looking at what's going on
on-chain.
838.7 --> 840.1: So there's a lot of other examples here.
840.4 --> 842.3: I want to jump to a slightly different world.
844.0 --> 849.2: And that is, you know, the world of hard-coded rug pull
s, rug pull scams.
849.4 --> 856.2: So essentially, we think about it as a market integrity
 element within the smart contracts themselves.
856.4 --> 859.3: At the end of the day, the assets are smart contract ba
sed on-chain.
860.2 --> 863.2: The way the contract is structured has a lot of impact
on integrity.
863.8 --> 868.9: I'm sorry, actually, I haven't updated the numbers here
 since the Georgetown block.
869.5 --> 883.6: But basically, you know, we define hard-coded rug pull
scams as essentially tokens that are issued, deployed specifically to sca
m people.
884.5 --> 898.0: And we've modeled over years of work, specific typologi
es where just by scanning the smart contracts, the moment it is deployed
on-chain, you know, there's certain pieces of code that there's absolutel
y no reason they would be there unless
898.0 --> 901.3: someone is, you know, engaged in foul play.
902.3 --> 908.8: You know, one very, sorry, one very common typology, fo
r example, is a honeypot, perhaps the most famous one.
910.3 --> 915.9: This is not the one I'm referring to, but the most famo
us one is Squid Games token for anyone who remembers that from a few year
s ago.
916.4 --> 920.3: Basically, when Squid Games was very, very popular, som
eone launched a token.
920.4 --> 922.1: That's often how these scammers work.
922.2 --> 925.3: They try to cling to something that people are interest
ed in.
925.3 --> 929.2: It could be, for example, a token issued by a large fin
ancial institution.
929.6 --> 934.6: So there are a ton of Biddle tokens, Biddle imposter to
kens, right?
934.8 --> 936.9: It could be a reference to popular culture.
937.7 --> 939.6: Squid Games was a big thing then.
939.9 --> 941.2: They launched a Squid Game token.
941.7 --> 947.5: And one of the first lines in the smart contract was ba
sically, you can invest in this, you can deposit, you cannot withdraw.

948.1 --> 950.7: So what happens is people invest quickly.
953.0 --> 958.6: Exactly. People, there are a lot of people out there th
at are constantly looking for things that might go in price quickly.
958.7 --> 960.8: People invest, they cannot withdraw, the price goes up.
961.5 --> 963.7: Towards the end of a smart contract is another piece of
 code.
964.4 --> 975.5: And you can kind of see how this is really market integ
rity, fraud, cyber, all beginning to coalesce into one as part of the tra
nsformation that finance is going through.
977.0 --> 981.7: But basically, one of the last lines in the smart contr
act was essentially a backdoor.
982.7 --> 985.1: And within a very short period of time, they ran away.
985.2 --> 988.2: The moment it reached about three and a half million do
llars, they ran away with the money.
988.9 --> 993.0: Now, I'm going to share some scary numbers.
993.2 --> 994.4: I don't know if I have them on this deck.
995.1 --> 1010.4: But we published an extensive report on it a few years
 ago where we found that one in four new tokens on the top 14 EVM chains
and Solana is a hard-coded scam.
1011.1 --> 1016.7: So you're also starting to see a reality of like kind
 of like what's, you know, the difference between scam and spam.
1017.4 --> 1028.5: I have this chart here, which just gives a sense of t
he deviation of these sort of assets of hard-coded smart contract scams w
ithin some of the largest ecosystems.
1029.5 --> 1040.2: You can see that after BASE was launched, it sort of
stole the, you know, it became the leading hive for this sort of activity.
1040.5 --> 1046.5: Because, I mean, we assume that it has in part to do
with the credibility that BASE brings into it.
1046.7 --> 1049.2: Before that, BSE was definitely the lead.
1050.2 --> 1052.2: And also, a lot of it has to do with cost.
1052.5 --> 1056.3: So if listing fees are low, that's oftentimes where f
olks go.
1057.6 --> 1064.1: Now, I want to point out, so we actually offer a webs
ite, tokensniffer.com.
1064.7 --> 1067.1: It's really the only product we offer directly to con
sumers.
1067.3 --> 1069.4: It's for free, so we don't make any money off of it.
1069.4 --> 1075.1: But really, if you go there, you'll see there's liter
ally a place you can plug in as any smart contract address and you'll get
 the report on it.
1075.1 --> 1082.2: And we scan, more or less in real time, all of the ne
w smart contract tokens, again, as I mentioned, on the 14 top EVMs and So
lana.
1083.2 --> 1086.5: So we just believe that ultimately transparency is gr
eat.
1087.0 --> 1088.8: It does not necessarily mean safety.
1089.7 --> 1095.2: I don't read code. Most people who will be engaged wi

th tokenized assets, digital assets, don't read code.
1095.5 --> 1098.5: The ability to get a quick sense of whether you shoul
d be careful is important.
1099.1 --> 1100.7: And that's kind of why we offer that.
1100.7 --> 1109.5: I do want to emphasize both in relation to hard-coded
 rug pull scams and in relation to the insider trading example I gave ear
lier.
1110.0 --> 1113.0: I said those numbers are scary, but there's a bit of
a silver lining.
1114.4 --> 1123.6: I'm optimistic because both of these demonstrate that
 when financial activity happens on-chain, you can automatically detect i
nsider trading.
1123.8 --> 1129.1: You can automatically detect fraud, which is not some
thing that traditional finance is able to do.
1129.8 --> 1132.0: Obviously, fraud happens in traditional finance as we
ll.
1132.0 --> 1137.2: I don't need to tell anyone how often scam calls happ
en, not even just in finance.
1138.0 --> 1144.2: Enough people have to get scammed and enough reports
have to be filed to enough regulators, usually, so that something will be
 done about it.
1144.8 --> 1154.4: You can start to see a reality being formed where usi
ng on-chain activity, you can start flagging things automatically for peo
ple in a way that wasn't available in the past.
1154.4 --> 1163.3: When you think about insider trading, most people tha
t I shared this data with are a bit shocked with the 56 percent number, b
ut they're not really that surprised.
1164.2 --> 1168.1: Naturally, permissionless ecosystems invite wrongdoer
s.
1168.6 --> 1172.5: But again, I want to reference that study I mentioned
 earlier.
1172.7 --> 1177.1: It still happens in the heavily regulated, highly con
stricted securities markets.
1177.6 --> 1183.9: Now, here, it's a new market, increasingly regulated,
 but still regulators are catching up.
1183.9 --> 1190.7: You can expect the numbers to be very, very high, but
 I also think that it's very likely that because it's much easier to dete
ct, it is higher.
1190.9 --> 1196.1: It's very possible that there's a lot more insider tr
ading happening that never gets detected in traditional finance.
1196.8 --> 1198.2: So again, there's a ton here.
1198.8 --> 1203.2: You know, we recently published research on meme coin
s and specifically Solana.
1204.0 --> 1216.6: We looked at pump.fun and found that based on our def
inition of a pump and dump, some of which is listed here, and by the way,
 all of this research is available on our website.
1216.7 --> 1219.7: You can download it or just reach out to me and I'll

send it to you.
1220.6 --> 1227.1: 98 percent plus, almost 99 percent of the tokens list
ed on pump.fun were rug pulls.
1228.1 --> 1231.5: Of course, they had a different, you know, we publish
ed it.
1231.5 --> 1236.1: Their response was, you really don't understand how m
eme coins work if you think those are rug pulls.
1236.4 --> 1241.6: But it's a bit of a question of like, you know, how y
ou define what kind of exchange you're trying to build.
1241.7 --> 1245.7: Is it, you know, is it a casino or is it actual finan
cial activity?
1246.4 --> 1256.0: Either way, a reality where such a high percentage of
 assets listed on a platform, even if you're OK with the fact that it's a
 bit more of a casino than a financial market,
1257.7 --> 1272.6: where so many assets, 98.6 of the assets there, you k
now, drop to liquidity of under $1,000 within hours, sometimes minutes, i
ndicates a very inefficient market.
1274.2 --> 1275.5: So that's interesting.
1276.4 --> 1285.2: I'm going to, I want to, again, by the way, if anyone
 is interested or if you'd like me to dive into anything more specificall
y, I just don't want to take too much time.
1285.2 --> 1287.4: So I'll kind of finish with wash trading.
1288.5 --> 1295.5: You know, we published this research also, and I'm so
rry that this slide is just a screenshot, but again, it's available.
1296.5 --> 1308.4: A couple of years ago, there was a lot of talk about
the fact that wash trading is not going to be a problem in decentralized
exchanges because it's expensive, you know, gas fees, etc.
1309.4 --> 1311.7: The transparency is there so you can see when it's ha
ppening.
1311.7 --> 1319.8: Again, I'm going to, you know, starring the point tra
nsparency because transparency does not mean that everyone understands wh
at the transparency shows.
1320.6 --> 1328.4: We looked at around 30,000 liquidity pools on Uniswap.
1330.0 --> 1336.8: And we, you know, the methodology is listed in a very
 detailed manner in the report.
1337.0 --> 1338.2: I won't dive too deep into it.
1338.2 --> 1349.1: But the bottom line is that we tried to, you know, we,
 so obviously there's a bit of the question of how do you define wash tra
ding in a swap based liquidity pool.
1349.2 --> 1354.5: In the first place, wash trading is a term that was,
you know, designed for centralized order books.
1354.5 --> 1367.5: But basically, we looked at liquidity pools where 90%
 of the activity, 90% of the liquidity was provided by, you know, one wal
let or a set of connected wallets.
1367.6 --> 1370.2: Sometimes they're very complex sets of connected wall
ets, etc.
1372.3 --> 1379.0: And the threshold we set, if I remember correctly, is

 50% of the swaps was also with wallets related to that wallet.
1379.5 --> 1381.0: That does not look like real trading.
1381.1 --> 1383.0: Now, of course, there's the question of why would the
y do it?
1383.0 --> 1383.6: It is costly.
1384.5 --> 1388.8: You know, wash trading is often very closely affiliat
ed with money laundering.
1389.3 --> 1393.3: It's a very good way of making money disappear or mak
e it harder to find.
1394.6 --> 1397.4: Sorry. Tax harvesting, of course.
1397.7 --> 1407.5: And also, oftentimes, interestingly enough, this sort
 of behavior is closely related to the sort of hard-coded rug pull scams
that we mentioned earlier.
1407.7 --> 1412.0: So you list a rug, you know, you list a hard-coded ru
g pull.
1412.5 --> 1413.8: You want people to invest in it.
1413.8 --> 1415.1: You want there to be activities.
1415.2 --> 1419.1: So you invest some gas fees in also creating that act
ivity.
1419.7 --> 1424.5: Anyway, and, you know, there, as I said, some of them
 are extremely simple.
1425.1 --> 1428.7: You know, A2A wash trading, the same actor essentiall
y trading against themselves.
1428.8 --> 1429.8: Obviously, that's very obvious.
1430.3 --> 1440.4: There are more sophisticated actors who, you know, en
gage in multi-wash, multi-party wash trading where there's a lot of very
sophisticated attempts to kind of siphon the money.
1440.4 --> 1449.5: And, of course, we're very glad to be working with ou
r KYT partners, you know, to be able to trace that sort of stuff.
1450.1 --> 1458.1: The last thing I'll throw in here, I'm not sure actua
lly how interested people are in here, but everyone in America right now
are talking about prediction markets.
1459.3 --> 1467.3: Now, obviously, it's important to say Kalshi, which w
e're very proud to have as a client, is the most famous prediction market.
1467.5 --> 1469.6: It's actually not a digital assets exchange at all.
1469.6 --> 1471.8: It's a derivatives exchange fully regulated by the CF
TC.
1472.8 --> 1476.6: But, you know, PolyMarket and a lot of other actors a
re fully on-chain prediction markets.
1477.6 --> 1489.3: And I'll be honest with you, I actually haven't seen
a lot of people interested in this outside of the U.S., you know, but it'
s a really big deal there.
1489.5 --> 1493.3: And, you know, I think it's fair to anticipate it's a
 really fascinating asset class.
1493.3 --> 1501.7: There's still a lot of debate on whether how much val
ue does it have, you know, beyond just the betting elements.
1502.3 --> 1504.2: But it's definitely here to stay.

1504.9 --> 1518.5: Obviously, there's a strong sports element, but also just, you know, just putting a price on information and creating a bit of like a brain hive ability to understand what people think.
1518.5 --> 1528.4: I don't need to tell you about all the fact that CNN uses, you know, Kalshi contracts sometimes as a sort of like counter to political surveys, et cetera.
1532.1 --> 1540.9: Now, prediction markets create a lot of new challenges when it comes to market integrity because you can essentially trade on anything.
1541.6 --> 1547.2: And when you trade on anything, there's a lot of new surface and vectors of potential insider information.
1547.7 --> 1548.8: I don't know.
1549.0 --> 1553.4: There was just, needless to say, there's a lot going on in Iran right now, in my home country, Israel.
1554.8 --> 1563.4: Someone could potentially be living next to a neighbor who's a pilot, and he can potentially notice that that guy has been away for three months.
1564.6 --> 1572.7: So you can potentially think, OK, I'll bet that there's going to be, I'll create a contract that there's going to be an attack somewhere.
1573.5 --> 1576.4: Is that insider trading or is it just very smart observation?
1577.0 --> 1578.2: Where does the line get crossed?
1578.4 --> 1594.4: And these questions are becoming more and more complex and sophisticated as prediction markets bring essentially any issue in the world into a traditional derivatives and futures reality.
1595.3 --> 1599.7: This is research that we haven't published yet, but we will soon.
1600.7 --> 1609.6: There's already a lot of research and media reports that were published about poly market and wash trading and other issues on its head leading to the U.S. elections.
1610.3 --> 1612.0: We went a few layers deeper.
1612.5 --> 1615.4: A lot of the research that was published was a little bit surface level.
1616.5 --> 1623.3: But we looked at 7.18 million trade execution rows in 27 days leading to the 2024 elections.
1623.3 --> 1632.6: And bottom line, we found that about $250 million worth of wash trading happens, approximately 14 percent.
1632.9 --> 1635.9: That number, by the way, is lower than some of the former estimates.
1636.1 --> 1644.2: Obviously, as you further methodologize and apply stricter rules and methodology, then the results are slightly different.
1645.0 --> 1649.1: So $253 million out of $187 billion.
1649.1 --> 1654.2: And by the way, that's on a few contracts related to the elections.
1654.9 --> 1669.9: One thing that was very interesting here, and again, just because we are focused on building a new lexicon for market manipula

tion and other forms of illicit activity on chain, is this idea that we d
efine as cross symbol wash trading.
1670.4 --> 1671.9: A novel form of wash trading.
1672.4 --> 1678.1: So, you know, a big chunk of the wash trading that we
've identified here is A to B to A.
1678.1 --> 1682.5: So essentially, two parties wash trading one against
the other.
1682.6 --> 1687.6: And obviously, you know, we have to think about the w
ay prediction markets work.
1687.6 --> 1691.9: They're a contract. So essentially, the price is alwa
ys $1.
1692.4 --> 1696.4: And then, you know, based on how many people buy each
 side of it, it's a binary contract, really.
1696.8 --> 1717.0: Then, you know, if more people buy, if more people, y
ou know, trade on the contract, on the contract that the, let's say, a ce
rtain candidate would lose, then the price, you know, goes up to buy that
 and the price on the other side goes down, right?
1717.1 --> 1720.8: And then, of course, you can make a lot of money if y
ou bet on something that was unexpected.
1721.4 --> 1725.1: If you not bet, sorry, it's really important to not u
se the term bet in relation to prediction markets.
1725.1 --> 1725.9: It's trading.
1727.0 --> 1732.6: And obviously, there's a lot of legal discussion goin
g on in the U.S. right now on where does the line get crossed.
1732.9 --> 1738.6: But this is on polymarket, the global on-chain.
1738.7 --> 1740.1: So it's not a regulated market.
1740.4 --> 1742.3: So, you know, you can choose whatever term you'd like.
1743.8 --> 1747.1: So, you know, ABA is essentially two actors that are
clearly working together.
1748.1 --> 1756.6: Cross-symbol wash trading is something we are really
trying to define as a novel form of prediction market specific manipulati
on.
1757.6 --> 1772.5: And essentially, in those cases, you'll see wash trad
ing that happens across multiple symbols that affect one another, right?
1772.5 --> 1792.7: Because sometimes part of what's crazy about predicti
on market, or let's say fascinating, maybe the word crazy is not right, i
s the fact that people, you know, you can sometimes impact the ultimate r
esult or the odds of a specific contract by creating other contracts.
1793.2 --> 1801.5: People essentially start treating the contracts thems
elves as information based on which they make decisions investing in othe
r things.
1801.5 --> 1821.8: So that creates a whole new vector for potential wash
 trading, not only between two actors that are essentially doing non-econ
omic trading, but those actors also doing it across multiple contracts, w
hich makes it more difficult, as you can imagine, to detect.
1822.5 --> 1824.0: Anyway, I'll stop there.
1824.3 --> 1825.4: I hope this was helpful.

1826.1 --> 1826.9: Yes, of course.
1830.2 --> 1833.4: Yeah, that was awesome.
1833.5 --> 1833.9: Thank you.
1834.1 --> 1844.0: I've written a lot down just to put on my meme coin trader, I'm in the trenches right now.
1845.0 --> 1849.1: One man's wash trading, another man's information asymmetry.
1849.7 --> 1851.3: And I think you kind of said that at the end.
1852.3 --> 1854.3: Wash trading is my launch strategy.
1854.6 --> 1855.5: What are you talking about?
1855.7 --> 1856.8: I need to get liquidity in the market.
1856.8 --> 1859.8: I need to show that volume's there.
1860.3 --> 1865.1: I'm going to set up a bot and buy my own token for a period of time and do that.
1865.5 --> 1866.6: And that happens regularly.
1866.9 --> 1877.1: Like that happens in almost every Solana meme coin, is that the person who launches the meme coin sets up a bot and sets up a pace to the buy and the market.
1879.4 --> 1880.4: What else did I write?
1880.4 --> 1884.1: I really like this scam versus spam situation.
1884.4 --> 1889.2: And at what point does something go from being a scam to just being spam?
1889.7 --> 1894.1: And is it about like how well you know that is a scam?
1894.2 --> 1897.2: Therefore, you're now just getting spammed when you experience it?
1897.9 --> 1898.8: That was interesting.
1899.7 --> 1903.8: Just on that point, I always try to think about it in the context of email.
1905.8 --> 1913.2: Imagine if when email was new, 98.6% of emails were spam, right?
1915.3 --> 1916.3: I'm saying scam versus spam.
1916.4 --> 1921.8: I mean, obviously, most of this activity we are interested in are actual scams, people trying to defraud other people.
1922.6 --> 1928.3: But there's also just a lot of empty activity on a lot of some of these platforms.
1928.3 --> 1943.2: And the bottom line is that if 99% of email activity before there were standards that were set in order to prevent scams, etc., was something between a scam and a spam, it's very likely we wouldn't have been using emails today.
1943.8 --> 1948.4: So it's a really core issue for these assets to be usable.
1950.6 --> 1957.6: And especially as they're further integrated into financial activity, as we bring more real world assets on chain, etc.
1960.9 --> 1962.3: Sorry, I have a couple of questions.
1962.8 --> 1968.4: First of all, when doing the rug pull research and wash trading research, did you look at Tron at all?

1970.2 --> 1975.1: Well, in this particular case, we looked, as I mentio
ned, like what we published, we looked at Uniswap.
1975.1 --> 1975.3: OK.
1977.8 --> 1981.0: Yeah, the DEX wash trading report was about Uniswap.
1981.6 --> 1983.8: We do have team members that have looked at Tron.
1983.9 --> 1987.4: I don't have any data on me, but happy to check.
1987.4 --> 1999.1: OK, yeah, I'm curious about obviously, I guess it's n
o secret that there's a lot of crime happening on Tron, but I'm wondering
 if that also extends to kind of market abuse related crime.
2000.1 --> 2011.3: Another question I have, and there may not be an answ
er to this, but considering that in certain jurisdictions, market abuse r
egulations don't necessarily apply to crypto.
2012.2 --> 2021.0: So, for example, in the UK, I think they're working o
n it or they were last I checked, but for a long time, at least it didn't.
2021.7 --> 2033.7: How in a cross-border world like crypto do you keep m
arket participants safe when there are multiple jurisdictions that wouldn
't necessarily cover that under regulation?
2034.7 --> 2042.2: I mean, I think you hit the nail on the head in the s
ense that it's one of the biggest regulatory challenges, the cross-border
 reality.
2044.6 --> 2051.8: And, you know, it's also really unprecedented in a lo
t of ways, although, you know, traditional capital markets are also extre
mely international.
2053.0 --> 2058.0: I think, you know, you're seeing the movement towards,
 you know, more regulation.
2058.1 --> 2063.0: I think it's also important to say that in traditiona
l finance as well, there are more and less regulated markets.
2063.7 --> 2065.6: You know, you can go invest in penny stocks.
2066.1 --> 2071.2: You can go invest in, you know, yes, exactly, bonds.
2071.7 --> 2076.4: Right. Generally, you know, liquidity flows to safety.
2077.8 --> 2085.7: So part of it is also just, you know, the market prov
iding these kind of two options, more regulated, more safe versus less re
gulated.
2085.9 --> 2087.7: And people can decide where they want to put their mo
ney.
2087.7 --> 2095.3: I mean, my bet is that ultimately most institutional
money and kind of mainstream money is going to go to where it's safe.
2095.6 --> 2107.1: That being said, I mean, look, we're going through th
is process of introducing more regulation and how increasingly in places
that introduce regulation early, updating the regulation kind of country
by country.
2107.1 --> 2117.8: But there's also a growing global effort where member
s of GBBC and I know, like Sandra, you're deeply involved with the OECD,
et cetera, and kind of harmonizing that globally.
2117.9 --> 2121.2: It's going to take time. I do think it's happening fa
ster than we realize.
2121.3 --> 2128.9: I kind of like to remind myself the I think the sayin

g is by Bill Gates, which I guess is no longer a very good person to refe rence.
2130.4 --> 2135.9: But but, you know, that we tend to over overestimate what can be done in a year and underestimate what can be done in 10.
2137.1 --> 2139.7: I mean, it's really kind of remarkable how fast it's happening.
2139.9 --> 2144.2: But so from a regulatory perspective, it takes time.
2144.4 --> 2146.1: Regulators are collaborating, they're improving.
2146.6 --> 2150.1: They're also really interesting cross-border regimes at this point.
2150.1 --> 2154.0: Like, for example, MECA in Europe, it's a true cross-border effort.
2154.5 --> 2161.0: We're very proud to work to be working with ESMA, hav ing me selected in their RFP.
2161.0 --> 2167.5: And it's challenging to understand how you even do it within the context of, you know, 30 countries that are unionized.
2167.6 --> 2168.8: But there's immense progress.
2169.5 --> 2172.9: And I think over time, we'll also see it on a global level.
2173.4 --> 2177.1: But ultimately, you know, we it's also up to us, righ t?
2177.1 --> 2181.5: It's up to organizations like Begin, which are, you k now, kind of building standards.
2181.6 --> 2190.2: And it's also up to the industry, because I mean, we' ve always believed that if we don't like the digital assets, tokenized as sets are not going to fulfill their
2190.2 --> 2193.0: potential if the markets are inefficient and unfair.
2193.5 --> 2199.9: And it's more likely that, you know, it's, you know, and every time you trade something, someone skims a little off the bottom and off the top, right?
2199.9 --> 2209.6: So you're seeing a lot of collaboration, like a lot o f really amazing industry association work that brings together a lot of platforms that are trying to standardize together.
2210.4 --> 2216.0: Naturally, it starts sort of like a platform by platf orm basis, and gradually it grows.
2216.8 --> 2220.4: And gradually, like, you know, umbrella activity is d eveloped there.
2220.5 --> 2232.5: One thing that I've always really appreciated about J apan specifically, is the JVCA's role and the idea of shared surveillance that, you know, has been discussed for a very long time.
2232.6 --> 2233.4: It's very complicated to do.
2233.4 --> 2246.5: But essentially, the fact that you need to share info rmation, and, you know, that I believe that over time, it will also lead to real time, cross market, you know, surveillance.
2246.7 --> 2248.6: I, by the way, hate, I don't love the term surveillan ce.
2248.7 --> 2251.0: I use it because that's what's used in traditional fi

nance.
2251.0 --> 2261.2: But essentially, yes, maybe observation, but essentia
lly, you know, use the advantage of the on-chain trading and the fact tha
t it's transparent.
2262.4 --> 2267.0: The number 20 percent of trading activity is on-chain
 is probably going to grow.
2267.9 --> 2276.6: Right. And then also find ways to share information a
round the off-chain pieces, which, you know, we launched something called
 the Crypto Market Integrity Coalition 2022.
2278.2 --> 2287.4: That essentially, you know, was the goal was to start
 by kind of pledging commitment to trade surveillance, but also over time
 developed ways for leading actors to share information.
2287.6 --> 2301.5: And I don't think there is ever going to be, you know,
 a reality where all trading activity is kind of, what was the term used,
 observed, or, you know, all trading activity and trading data is, you kn
ow, kind of pulled together.
2301.5 --> 2303.6: So you can really understand what's going on in the e
ntire market.
2303.6 --> 2311.1: But I believe that leading platforms are going to sta
rt to creating slightly more, you know, are going to start to collaborate.
2311.4 --> 2313.7: And then it's going to be a bit of a line in the sand.
2313.9 --> 2321.4: Like, you know, you'd want to work with the platforms
 that are also collaborating to protect cross-border, et cetera, versus p
latforms that don't.
2321.5 --> 2329.3: So in that sense, the market, you know, market and de
mand and just, you know, where people prefer to put their money is, I thi
nk, also going to help solve some of it.
2329.7 --> 2331.6: Sorry, very, very long answer to a simple question.
2331.6 --> 2332.4: Thank you.
2333.2 --> 2335.0: Can I ask one question?
2335.4 --> 2337.0: Thank you very much for a great presentation.
2337.1 --> 2340.4: It's very good education for everyone about what happ
ened.
2342.5 --> 2348.3: My question is very simple, but I would like to expla
in my background on my question first.
2349.2 --> 2355.5: This project, you know, that creating some dictionary
 or dictionary for the illicit activities.
2355.5 --> 2363.0: So this project started two years ago at the book num
ber 12, 10, two years ago.
2363.7 --> 2373.8: Then the reason why we started that discussion was th
at, you know, there, you know, regulators say that this is an illicit act
ivity or something like that.
2373.8 --> 2386.0: But, you know, the creator of technology, like engine
ers, you know, and the new start-ups doesn't understand that what, you kn
ow, regulators say, this is an illicit activity, this is not.
2386.7 --> 2395.7: So at that moment, I felt that, you know, the languag
e around illicit activities are very different between the regulators and

engineers or start-ups.
2396.2 --> 2403.0: So this is a strong reason why I felt that we need to create some kind of common dictionary to fill the gap of understanding.
2403.8 --> 2412.9: But at this moment, so after hearing that presentation, so, you know, the details of the illicit activities or harmful activities are very complicated.
2413.7 --> 2422.7: So it's not, so I felt it is not easy to describe the details of each illicit activities in natural language.
2425.1 --> 2444.8: But, so, on the other hand, so if we can, you know, mathematically form our expression about the illicit activities, so it might be interesting direction because, you know, so Mitchell is creating that AI-based, you know, some, you know, knowledge base about what's happening in the space of the crypt.
2445.4 --> 2453.0: So if we have some mathematical form of expression, it might be a programming code, it might be a mathematics, but it might be equation.
2453.5 --> 2461.1: But anyway, so we can have some special style of expression about what the illicit activity looks like.
2462.1 --> 2470.1: And, you know, we can utilize AI system or some computer system, and, you know, we grab some, you know, list of past transactions.
2471.0 --> 2479.2: It's kind of a data set, and we can signaling that, you know, for every, you know, general crypt users.
2479.6 --> 2491.6: So it might be, you know, scam or something like that, but it's kind of a good way to think about, you know, having the formal description rather than the natural language description about the illicit activity.
2491.9 --> 2496.2: This is my, it's kind of a stupid idea, but it might be an interesting idea.
2496.3 --> 2497.1: What do you think about that?
2497.5 --> 2500.1: I mean, I don't think, I don't think it's silly at all.
2500.3 --> 2514.8: I think that's, I think jargon has had a very, like overly complex jargon has had a negative effect on this industry for a very long time.
2515.1 --> 2515.8: It's also natural.
2516.0 --> 2516.6: It's very innovative.
2516.8 --> 2524.4: There are a lot of ideas coming up all the time, but oftentimes, you know, things are described in ways that are more complex than they really are.
2524.4 --> 2535.6: I mean, like when I speak with folks who are not from the industry about smart contracts, for example, sometimes it's easier to explain that they're just a set of, they're just code really, right?
2535.7 --> 2544.2: Like, you know, smart contract is, you know, is an amazing term, but, you know, if you want to explain to someone, you know, something, it's a set of pieces of code, right?
2545.6 --> 2547.6: Much, much easier to understand what it is, right?

2547.6 --> 2560.4: If you say, like, I think today is also different than five years ago, but five years ago, you'd go to, I don't know, you know, some traditional institutions, et cetera, and you'd say a smart contract, and it sounds really cool, but like, it's also, what is it, right?
2560.6 --> 2564.7: So I actually think, I think, I don't think, I think it's a, it's a great idea.
2565.3 --> 2583.4: I think part of the challenge there is also the point you raised when it comes to market, you know, money laundering, sanctions, those worlds of compliance are, you know, I wouldn't, they're obviously not simple, they're extremely complex, but there's a bit more of a binary reality.
2584.4 --> 2592.0: At the end of the day, when a specific actor is sanctioned, you've either touched that or you didn't, right?
2592.0 --> 2594.3: And if you did, you know, it's wrong.
2594.5 --> 2614.4: Market manipulation, as you said, one person's market manipulation is another person's very sophisticated trading, and ultimately, you know, society puts together a law that defines where the line is put, and then courts oftentimes need to decide whether it was crossed or not.
2616.6 --> 2642.8: So if I understand, Professor, you're suggesting correctly, yes, I think that being able to kind of dynamically, and based on popular opinion, or, you know, and utilizing AI, you know, defining and creating a more dynamic reality of like how you define these terms and how you add new terms, I think would be very, very valuable, especially if we're able to do it in a, you know, in a way that's simple enough for people who are not necessarily technical to understand.
2643.4 --> 2644.3: Thank you very much.
2644.5 --> 2645.9: So is that correct?
2646.1 --> 2654.4: And I felt that, you know, natural language is too naive to express the complicated situation happening now.
2656.3 --> 2657.8: Sorry, could you repeat that?
2658.1 --> 2658.6: Yeah, yeah.
2658.8 --> 2668.0: You know, natural, simple natural language that, you know, the scam or something like that, it's a very, you know, too simple to express that complicated situation.
2668.9 --> 2669.9: Yeah, I mean, true.
2669.9 --> 2682.4: By the way, it's actually, you know, I didn't go into it, but the role of agentic AI is becoming really critical in driving compliance, in my view, to a better place.
2683.9 --> 2691.0: You know, we have deployed a set of AI agents on our market integrity hub that our clients now use.
2691.5 --> 2698.4: Now, there's another issue with the world of market manipulation trade surveillance is there's a significant shortage of talent.
2699.9 --> 2720.6: You find a lot of, especially with specialty in digital assets, but also, by the way, you know, in legacy assets, there's, you know, it's, you find a lot of, you know, it's the day to day of trade su

rveillance professionals.
2720.8 --> 2726.4: Compliance professionals in general is, you know, I a
lways think about them as the unsung heroes of finance.
2726.7 --> 2729.1: They are fighting to protect other people's money.
2730.2 --> 2742.3: But, you know, it's full of false positives resolutio
n and very repetitive work and having to oftentimes, like, spend a lot of
 time drafting language to document what happened in really complex situa
tions.
2742.3 --> 2750.4: Like you're saying, you know, working with an agent t
hat doesn't necessarily do the detection, that's a problem, right?
2750.4 --> 2756.2: Because when it comes to detecting crime, you need sy
stems that are deterministic and that bring to the same results only.
2756.5 --> 2772.0: But on the remediation, on the documentation, on the
naming, that's where agents are extremely effective and already really, r
eally help compliance professionals by simply proposing a solution for wh
at happened that they can review versus then having to kind of come up wi
th the language immediately.
2772.9 --> 2780.1: So it allows compliance professionals to focus on wha
t they signed up for, which is fighting crime, not, you know, marking fal
se positives.
2780.4 --> 2781.7: Thank you very much.
2785.6 --> 2791.5: Thanks for, I mean, I've heard you speak many times,
but this was a great update to the great work.
2791.8 --> 2794.6: No, it's it's incredibly important, the work that you
 guys are doing.
2794.8 --> 2800.1: And I'm going to piggyback off of the agentic AI poin
t you raised, because that was actually my question.
2801.1 --> 2811.6: Doesn't the acceleration of agents and bots now being
 used in the DeFi space only going to compound and turbocharge the proble
ms that we already have?
2813.2 --> 2816.0: And, you know, it can be used for good and for bad, a
nd we all know that.
2816.1 --> 2820.4: But the point being is they are being deployed, parti
cularly in DeFi.
2820.4 --> 2833.4: And I don't know that, with all due respect to regula
tors and policymakers, I don't even think we're at the place where that i
s being discussed, let alone the more fundamental things that are being w
orked out.
2834.0 --> 2837.5: How much work are you guys looking in that space righ
t now as it evolves?
2839.2 --> 2843.4: Really, really important question and also very multi
layered.
2843.6 --> 2845.7: But I mean, I mean, I think you're right.
2845.7 --> 2852.4: And by the way, that's a problem naturally in our wor
ld, but it's also a problem on X, right?
2852.8 --> 2859.6: Like, you don't know what to believe anymore because
there's so much volume that is not necessarily real people and you can't

really distinguish.
2860.9 --> 2866.3: So when I say multilayered, so I think we sort of lik
e are increasingly graduating from the term bot to the term agent.
2866.9 --> 2869.9: And they're similar, but very different things.
2870.4 --> 2877.9: So if we have this issue that is not resolved with bo
ts, and now we're going into an era of agentic commerce, agentic finance,
 et cetera.
2878.0 --> 2885.5: Now, mind you, again, just kind of reminding myself a
bout jargon and how everyone gets excited about buzzy terms, like we're n
ot quite there yet.
2885.6 --> 2888.2: But needless to say, it seems like it's happening qui
ckly.
2889.3 --> 2897.7: And you're going to have these bots, agents, which ha
ve significantly more agency than a bot that was designed for one purpose.
2897.7 --> 2900.0: And that are doing things on behalf of people.
2901.0 --> 2904.6: Needless to say, like any technology, you know, crimi
nals are going to be using them very effectively.
2905.8 --> 2924.5: And I think that you're absolutely right that we are
still in, you know, the world of financial crime is still in very early s
tages in even grasping what it means to do compliance and financial crime
 in an agentic commerce space.
2924.5 --> 2932.1: I think one thing that is probably, you know, true is
 that the only way to deal with it is going to be with other agents.
2934.8 --> 2937.5: Now, again, it obviously introduces many new challeng
es.
2938.8 --> 2945.4: You know, this, but yeah, but I mean, you know, we're
 already not keeping up with relatively static bots.
2945.4 --> 2959.2: We're not, you know, imagine an agent that is given a
gency to go and, you know, deploy new bots constantly and agency to adapt
 them over time to run away.
2959.3 --> 2961.8: So, for example, I mentioned TokenSniff for our websi
te.
2963.0 --> 2966.0: We get a lot of the scammers themselves.
2966.5 --> 2973.2: They go on our website, they go on the website, they
test the contract before they, you know, before they deploy it, right?
2973.2 --> 2977.2: An agent can do that significantly more effectively.
2977.9 --> 2988.1: It's funny because they sometimes communicate with us
 because they really don't like folks trying to kind of combat it, but li
ke they'd literally put swear words in the smart contract code to kind of
 try and insult us.
2989.6 --> 2998.6: So, bottom line, it's a massive issue and it's likely
 that it's only going to be resolved with other agents and that opens a w
hole other world of questions.
3000.6 --> 3002.0: Could I jump in here?
3002.0 --> 3003.9: Yeah, of course. I'll just mention one thing.
3004.0 --> 3012.5: I mean, I think this point about deterministic versus
 probabilistic AI and where it's going to use in law enforcement and fina

ncial crime is critical.
3014.6 --> 3017.4: We define our approach as neuro-symbolic.
3017.7 --> 3028.4: So, basically, not every form of AI, you know, not every form of AI is fit for every purpose.
3029.3 --> 3037.6: Symbolic AI, you know, refers to the kind of AI that's been done since the 70s, like used in doctor's offices, et cetera, algorithmic yes-no trees, right?
3038.6 --> 3039.4: It's deterministic.
3039.5 --> 3041.8: You put the same information, it will give you the same result.
3042.6 --> 3057.9: That's generally, you know, we believe the right approach for the actual detection of market abuse because ultimately if you're kicking someone off an exchange, you want to be able to show that, you know, you want to be able to explain the logic and you want to be able to show
3057.9 --> 3060.9: that if another person does exactly the same thing, the same result would happen.
3061.3 --> 3065.5: LLMs, probabilistic models, notoriously can give different results.
3066.8 --> 3068.1: That's part of the idea, right?
3068.1 --> 3079.3: They generate new information, but that is, you know, obviously much more advanced and can do a lot of things that symbolic AI can't, but that's extremely effective.
3079.3 --> 3091.1: More on the remediation side, on working alongside humans and allowing humans to focus on the right places and also allowing the symbolic models to kind of be used only for the specific purposes they are.
3091.2 --> 3102.3: So anyway, there was a bit of a scattered answer, but I mean, I think it's one of the biggest questions that we all should be asking ourselves, even if we haven't yet resolved the questions that we're currently dealing with.
3102.7 --> 3102.9: Sorry.
3103.4 --> 3104.5: No, no, that's good.
3104.7 --> 3114.5: And I am going to tie it together with the prediction market conversation and some research I've been doing into poly market and it's been published as well.
3114.6 --> 3132.2: Not my research, someone else's research is that of the top 10 poly market traders, six of them are bots of like the second one is what you were describing before, where this bot actually only takes trades on poly market that are made impossible.
3133.2 --> 3135.5: After different things happen in the world.
3135.6 --> 3151.0: So there is like an arbitrage lag that AI can pick up on way quicker than especially in sports, when a trade that was made a market is no longer possible because another market is closed.
3152.1 --> 3158.0: And this bot, that's how it trades and it's literally one of the biggest winners on poly market.
3158.5 --> 3180.7: And I think that the reason for this like poly market

prediction market acceleration is because people are realizing that AI o r LLM specifically, not only can do this within the market, but they can, as you rightfully said, they can then deploy another 10 bots to then slo p on X and then shift the market.
3180.7 --> 3200.7: And if they have enough power and sway in those multi ple social media, they could choose to lose in order to win the bigger tr ade by making the losing trade, well, making the winning trade like the i mpossible win by manipulating their way to the losing trade through slop.
3200.7 --> 3203.5: And like, literally, I'm pretty sure I've seen this h appen.
3203.9 --> 3212.8: But yeah, I just find it interesting now that you get this massive variety of markets that can emerge.
3213.9 --> 3215.9: How do we manage that?
3216.0 --> 3217.0: I think it's an open question.
3217.5 --> 3218.0: But yeah.
3219.3 --> 3221.3: I mean, I agree that it's an open question.
3222.4 --> 3223.9: It's a really, really important one.
3223.9 --> 3224.8: And by the way, you're right.
3224.8 --> 3226.2: It happens all the time.
3226.2 --> 3234.7: In a market where there's absolutely no limitations a nd no effort, I mean, I'm not necessarily talking about poly market, I'm talking broadly.
3235.4 --> 3240.4: If there's no effort to try and curtail market manipu lation, then obviously there's going to be more of it.
3241.0 --> 3246.8: And, you know, agents provide a lot of new ways to do it more efficiently and in harder to detect ways.
3247.6 --> 3253.0: I guess not exactly responding to your point, but sor t of echoing it and also going back to Sandra's point.
3253.0 --> 3267.2: One thing that's really important in general, in term s of becoming more effective, detecting market manipulation, and specific ally in the context of utilizing AI, is obviously what data do you use?
3267.8 --> 3268.5: What data sources?
3268.7 --> 3282.6: And one thing that was really interesting for us as p art of our journey as we've built our system, built the company, started working with clients, etc., was that historically trade surveillance syst ems, the ones used for securities and derivatives,
3283.3 --> 3287.4: really only reference order book and market data.
3293.9 --> 3301.7: Now, the problem is that for that, you depend on very standardized data, you depend on very consistent data sources.
3303.2 --> 3306.2: And that was not the reality in digital assets.
3306.5 --> 3308.7: So we couldn't build our system that way.
3308.7 --> 3320.1: Instead, we started developing ways to use more sourc es of data, even if each data source is a little less complete, or this w ay you're not dependent on just one data source.
3320.7 --> 3335.0: Now, obviously, AI is much better than, you know, hum ans or just, you know, or even just, you know, traditional algorithms, if you want to call them that, at looking at many more kinds of data in one

place.
3335.0 --> 3354.6: One thing, for example, that's, you know, we think is really, really important to do trade surveillance properly, but ultimately leaks to a lot of other elements of financial crime, is, you know, asking the question, why are we looking at transactions in a completely fragmented manner than we're looking at trading data?
3354.6 --> 3383.7: You know, the FCA, in its AML guideline that was released January 2025, you know, it essentially had a lot of conversations and surveyed, it's, you know, not to do with digital assets, just other regulated firms, and pointed out that compliance departments that unified, or at least looked at transaction monitoring data, so AML related data, transactions, deposits, withdrawals,
3385.0 --> 3391.2: together, under one system, with trading data, were much more effective at doing both.
3391.8 --> 3406.5: I actually have an example that I forgot to show here that I think is really interesting, but just to say that, like, part of, you know, what AI is going to do is it going to allow us to do that much more effectively, and hopefully over time fight financial crime, you know, more effectively.
3406.5 --> 3419.9: There's an example, actually, that is very Japan specific that I showed, sorry, that I showed in Washington, that I guess I'll just reference, because it's, again, it's very Japan specific.
3420.4 --> 3426.3: So, you know, you know, I point, I was kind of, I hope it's okay, I share the screen again.
3427.3 --> 3446.8: You know, you know, I was showing a lot of new, sophisticated, nuanced, in some cases, weird forms of market manipulation that exist in digital assets, but naturally, you know, illicit actors are becoming more sophisticated, not just in digital assets, also in traditional finance.
3447.8 --> 3463.1: This was a big story on Bloomberg last year, essentially, you know, a really sophisticated scheme that combined two separate vectors, one of fraud and one of market manipulation, to be able to evade authorities here in Japan.
3463.1 --> 3485.0: I believe it was in penny stocks, but essentially, they, you know, the way they operated is they, first of all, created fully KYC'd, legitimate, funded, et cetera, accounts on trading platforms, and established long positions on various assets that had very low liquidity.
3486.3 --> 3487.7: That was one vector.
3488.3 --> 3492.0: On the other side, there was an account takeover ring.
3492.2 --> 3508.5: Now, needless to say, they used a lot of different ways, as hackers do, you know, social engineering, et cetera, cyber tactics, but they were able to take over accounts, generally accounts that were not very active.
3509.2 --> 3522.5: And once they were able to take over accounts, what they did is they placed trades on the same illiquid assets that they had long positions on in the other vector.
3523.1 --> 3526.7: So the prices started moving very, very quickly becau

se they were very illiquid assets.
3526.8 --> 3537.7: Now, on both sides of the aisle here, both vectors, t
here was nothing really that was going on that seemed wrong because they
were able to get access to the account.
3538.5 --> 3541.8: You know, there are obviously fraud prevention system
s to detect account takeovers.
3542.3 --> 3544.2: They didn't work, you know, in a lot of these cases.
3544.8 --> 3551.0: But the way account takeovers are often detected is i
f activity takes place, it is really, really unusual.
3551.5 --> 3560.6: So, you know, for example, classic account takeover,
you know, detection, red flag, funds are withdrawn very, very quickly.
3560.8 --> 3563.5: But those were not accounts that had a lot of funds i
n them.
3564.9 --> 3567.5: And, you know, and again, like they were able to take
 over the account.
3568.2 --> 3574.3: There were no red flags when looking just at the acco
unt activity, just at the transaction monitoring, just at the deposits or
 withdrawals.
3574.4 --> 3575.7: So they deposited some funds.
3576.0 --> 3579.2: They didn't like try to steal anything because that w
asn't the game here.
3579.4 --> 3584.4: Right. They instead were able to manipulate the marke
ts through those accounts.
3584.6 --> 3591.2: And then once the price jumped, going back to the fir
st vector, the other side of the aisle, they were able to take the profit
 again.
3591.2 --> 3595.8: And there were no red flags because those were fully
legitimate accounts that placed those long positions.
3595.8 --> 3597.5: And it's anyone's right to place a long position.
3598.0 --> 3603.3: And, you know, obviously, you know, we don't want to
live in a reality where everything is assumed to be manipulation.
3603.7 --> 3610.0: Right. Now, what was really interesting to us about s
o and it was it was a significant amount of money.
3610.0 --> 3615.8: And it took, if I remember correctly, from the report
ing, you know, a couple of years before it was detected.
3616.5 --> 3625.8: Now, it was interesting for us because that's a it's
a modus operandi that we see a lot in crypto.
3626.5 --> 3643.2: So as an example, you know, you see a and again, beca
use of the way crypto works and because of the fact that you need to cons
tantly kind of correlate what's happening on chain with what's happening
off chain on the account.
3643.9 --> 3653.2: You know, most of our clients, you know, crypto excha
nges, et cetera, actually use our system to look at both transaction moni
toring and market surveillance in one place.
3653.6 --> 3655.1: So we detect stuff like that a lot.
3655.6 --> 3659.2: Now, again, we needed to build it that way because of
 the way crypto works.

3659.3 --> 3662.8: I also understand why it wasn't designed that way in
traditional finance.
3663.6 --> 3676.0: But the bottom line is that we are, again, like hinds
ight 2020, but we're pretty convinced we're very confident that our syste
m would have been able to detect this or at least the system that was loo
king at both transactions at the same time.
3676.4 --> 3690.6: So I'm trying to bring it all back to agents and AI a
gain, like the ability to use AI to look at more data in one place to fin
d correlations that we wouldn't have been able to find otherwise.
3690.6 --> 3710.1: It doesn't necessarily mean that you need to hold tra
ding every time you found an anomaly, but, you know, to use it to underst
and where there are risk or something doesn't seem right is going to be r
eally, really critical in dealing with some of these new forms of manipul
ation, whether it's in crypto, whether it's agentic driven.
3710.5 --> 3714.4: But I believe also in making securities and derivativ
es markets safer over time.
3714.4 --> 3723.7: And again, I want to be an optimistic here, not like
not a scarecrow and say that I actually think there already is a lot.
3723.8 --> 3728.5: I mean, I don't think I know because we do work today
with companies that don't do digital assets at all.
3729.2 --> 3734.4: There's a lot the traditional finance can already lea
rn from crypto about fighting financial crime.
3734.5 --> 3737.4: It's, by the way, true as well, I think, on the side
of AML.
3737.4 --> 3755.9: But over time, I believe we'll see more data being pu
lled together in order to be able to be more effective at detecting these
things, especially in light of how sophisticated, you know, criminals ca
n use agents and just do things that we can't even imagine.
3758.2 --> 3764.7: I'm going to open the conversation up to AML, CTF, KY
C, threats and harm.
3764.8 --> 3776.6: So on that side of the common lexicon, too, as well a
s pause for a moment for anyone online who had something that they wanted
to add to the conversation.
3777.4 --> 3784.1: Feel free to provide like a reaction or put your hand
up and we can make sure that we get to you.
3784.9 --> 3791.6: While we wait for that to potentially happen, I just,
oh, yeah, OK, I see.
3792.7 --> 3796.9: So thank you very much for the brilliant presentation,
as always.
3797.0 --> 3799.1: So I'm one of the regulators.
3799.3 --> 3803.0: So let me share some observation and let me ask one q
uestion.
3803.7 --> 3810.9: So I'm taking care of AML safety and financial stabil
ity of this asset, but not market surveillance.
3810.9 --> 3835.0: Fortunately, in a sense, because I guess market surve
illance is much more challenging for regulators and the market inspectors,
because, let's say, in the AML safety world, you know, if we identify a

kind of suspicious wallet, you know, we can freeze or burn those accounts,
 even though we do not know real identity of the holders.
3835.0 --> 3846.6: But when it comes to the market manipulation, like in
sider trading, we have to kind of, you know, show, kind of, you know, pro
ve that, you know, this guy is actually doing the wrong thing.
3846.7 --> 3852.3: So I think that that hurdle is much higher, potential
ly, than the AML safety cases.
3853.0 --> 3859.7: So as some of you might know, we are going to enhance
 market regulation.
3859.7 --> 3878.2: And if the bill that we propose to the parliament wil
l be approved this year, then my colleagues at the FSA and the Japan Secu
rity Exchange Commission will have to kind of enhance and develop market
surveillance system going forward.
3878.2 --> 3906.0: But, you know, of course, we have a lot of talented p
eople, have good experiences and knowledge in the traditional financial m
arket, market surveillance, as well as, you know, crypto and AML safety,
but they're not, of course, you know, it's a kind of new challenge to dev
elop some knowledge and experiences and a system internally or relying on
 third party experts like Surya Swarbu.
3906.0 --> 3912.4: So I think my question is, you know, it's a very open
 question, but how we should do that?
3912.6 --> 3917.3: So, you know, what is the kind of, you know, practica
l step for regulators?
3917.6 --> 3931.0: You know, maybe we have to increase the number of peo
ple, but, you know, we need more education and then we have to internally
 develop a market, you know, manipulation detection system, or we should
rely on a third party like Surya Swarbu.
3931.0 --> 3938.0: Or, you know, we can get the data by ourselves and th
en we can rely on data vendors like Kaiko, for example.
3938.2 --> 3940.8: So that's kind of, you know, we have to think about a
 lot of things.
3941.1 --> 3945.8: But I'm glad if you could give us some advice on that.
3947.5 --> 3956.2: You know, I have so much sympathy for regulators beca
use obviously it's really like, you know, no one can keep up with everyth
ing that's going on.
3956.5 --> 3961.9: And regulators have a responsibility to do everything
 possible to protect people and make sure there are fair markets.
3963.1 --> 3966.1: I don't think my answer here would be particularly or
iginal.
3966.1 --> 3974.3: I think that, first of all, you know, the most effect
ive regulators are the ones that really engage with industry.
3974.8 --> 3981.7: And the education is mutual, like always, like, you k
now, every time we have a conversation with a regulator, we learn a lot.
3982.0 --> 3984.6: And, you know, I think that they also do.
3984.8 --> 3991.2: So, again, I think FSA specifically has been really,
really, really strong at that.
3991.5 --> 3995.1: And a lot of regulators, you know, kind of, you know,

caught up over time.
4000.0 --> 4006.8: You know, there's needless to say, training and education, what everyone's talking about, it was true five years ago, it's true today, it will probably be true forever because things will continue to change.
4007.4 --> 4010.3: You know, we try to do our best to provide resources.
4010.9 --> 4017.9: We launched something called Trade Surveillance Academy, which, you know, there's a symbolic fee.
4018.3 --> 4022.1: But if anyone here is interested, I'm happy to give a waiver, you know, mostly for seriousness.
4022.1 --> 4026.6: But, you know, really to provide training.
4028.8 --> 4042.8: You know, to your question on, you know, what's the best, like, you know, if I understood correctly your question, Riosuke, you know, should regulators kind of develop their own systems or work with the market?
4043.0 --> 4045.9: Like, I think often, obviously, I'm a little biased here.
4046.4 --> 4048.3: I think everyone should buy Solidus Labs.
4048.6 --> 4049.2: No, I'm kidding.
4049.2 --> 4051.9: But no, no, I'm joking.
4052.2 --> 4062.1: But I mean, I mean, by the way, this question is very true in, you know, working with private companies as well.
4062.1 --> 4064.0: There's always this build versus buy question.
4065.4 --> 4075.6: I find that oftentimes our partners that, you know, do best, whether a regulator or not, you know, have a little bit of both.
4075.6 --> 4081.7: But I will say that, you know, you know, specialization is pretty critical in our economy.
4081.8 --> 4090.9: Right. You know, we've had cases where companies invested a lot of money in building a really strong system, but then you also have to maintain it, etc.
4091.1 --> 4095.8: Right. There's kind of like a reason why companies like ours has business, which is exactly that.
4096.8 --> 4101.1: Yeah, but I mean, I think that, you know, strong engagement, investment in education.
4103.0 --> 4130.5: I also think, by the way, it's a bit, it's not just for regulators, but for anyone, I think, again, I point out to this jargon problem, like, I think there was a lot of time where a lot of people were really, not just regulators, anyone, including me, were really concerned about sounding silly, asking questions that might sound stupid, but then things move so fast in this industry that if you don't, you know, if you're not going to open about the fact that you don't fully understand something, then it's really easy to, you know, to get, to be left behind.
4131.1 --> 4138.3: Yeah, but I mean, maybe I'll summarize all of this by saying private-public partnerships are key.
4139.0 --> 4141.3: Again, something that GBBC has been really championing.
4141.4 --> 4143.8: I'm sorry, Sandra, I keep on referencing you, but I h

ope it's OK.
4145.2 --> 4150.7: Yeah, but private partnerships, you know, private-pub
lic partnerships are key.
4151.5 --> 4164.8: We really make a point to, obviously, you know, to tr
y and be a resource for anyone who's interested and create resources, eve
n when they're not necessarily commercially viable, like, for example, th
e Trade Surveillance Academy.
4166.2 --> 4168.5: Yeah, we need to continue having these conversations.
4170.2 --> 4176.4: And I think that over time, also, adoption of AI is g
oing to help kind of reduce some of the workload.
4176.4 --> 4183.7: The way we think about AI in compliance, and I think
the same is true for supervision, is not about replacing people.
4183.9 --> 4188.2: There's already, like, there's not enough, you know,
compliance professionals as is.
4188.6 --> 4192.1: It's really about allowing humans to focus on the mos
t important stuff.
4192.2 --> 4200.9: And I think that, again, because so much of the work
is so repetitive, so today in compliance, and I assume also in supervisio
n.
4201.5 --> 4204.0: So I think that also is going to, you know, make thin
gs a little bit easier.
4205.0 --> 4218.5: Yeah, and just, you know, continue having that virtuo
us cycle of engagement, education, when appropriate, deploying technology,
 you know, that's fit for purpose.
4220.0 --> 4226.9: And, you know, whether it's, you know, bringing in a
third-party vendor or increasing the usage of AI.
4228.1 --> 4235.8: But, again, it's important to remember, like, it's no
t like the situation today with financial crime prevention is that glorio
us, right?
4235.9 --> 4237.3: Like, it's not resolved well.
4237.4 --> 4239.4: So I actually think we can only go up from here.
4240.8 --> 4247.0: And that dynamic between industry and government is c
ritical.
4247.7 --> 4248.6: I hope I answered your question.
4248.7 --> 4249.7: I feel like I was a little bit scattered.
4253.8 --> 4260.5: One more thing, and by the way, I completely agree wi
th you on the public-private information sharing and partnership.
4261.0 --> 4271.7: But I think one thing that it's going to be required,
 which is quite challenging, is the time of being able to recognize somet
hing has happened that is bad, and then reacting.
4272.4 --> 4278.6: And I think waiting two years to find that there's a
problem is not going to be acceptable anymore in a digital world.
4278.6 --> 4294.4: So then the question will be is what needs to happen,
 what structural changes need to happen on the regulatory side with indus
try in order to help accelerate the response time, and then ultimately th
e enforcement of whatever bad things have happened.
4294.8 --> 4304.7: And I think one place where I feel like we've been ta

lking about this amongst some of the industry partners is when something happens in a jurisdiction,
4305.4 --> 4314.5: those use cases need to then be put into a database or some kind of global catalog so that everyone can learn from what happened.
4314.9 --> 4323.8: I don't know if anyone's seen in the last, what is it, a couple weeks, South Korea has had their tax authority and the police department have made some major blunders.
4323.9 --> 4327.8: I mean, some of it's pretty basic, but we all need to learn from it.
4327.8 --> 4342.8: And not hoping this never happens with JFSA, but if other regulators also have issues or there are specific cases where you could learn to prevent those things from happening, to me that is just as valuable as the other stuff that's needed.
4343.1 --> 4350.9: And we are not doing enough information sharing, and I'm sorry, but going to X for all your information is probably not going to be as helpful as it should be.
4351.4 --> 4355.2: This is what we are trying to create at this project.
4357.8 --> 4358.3: Yeah, completely.
4359.4 --> 4361.1: Yeah, yeah, 100%.
4361.1 --> 4362.5: And it also, it fits.
4363.2 --> 4366.5: What begins creating, so from the global point of view, borderless point of view.
4367.0 --> 4371.2: And it fits into the cybersecurity threat and incident response work as well.
4371.6 --> 4383.2: So the cybersecurity threat and incident response database is the more securitized version of the common lexicon that we can establish here.
4384.2 --> 4387.2: I did have one question for you, Professor Matsu.
4388.6 --> 4398.0: The mathematical expression of market manipulation makes sense, but is it possible to get to a mathematical expression of an AML compliance?
4398.4 --> 4399.8: Yeah, that is a research question.
4400.1 --> 4400.3: Okay.
4400.5 --> 4405.8: But of course, it's not perfect, but so, you know, creating some model could help, yeah.
4406.1 --> 4408.2: Because I like the idea of coming up with this.
4408.2 --> 4410.6: Could help for the automated prediction or automated...
4410.6 --> 4412.0: Yeah, signaling or something like that.
4412.0 --> 4416.9: You create like a compression of like, this is what this particular scam looks like mathematically.
4417.0 --> 4421.9: And then you give that to the AI and it's just like, look at all the data.
4422.0 --> 4427.0: And if any of this math matches, we can get this like surface area almost.
4427.3 --> 4434.4: It's not legal evaluation, but it's kind of the predi

ction or it's kind of the signaling process.
4434.4 --> 4434.9: Yeah.
4435.9 --> 4436.4: Yeah.
4436.7 --> 4442.1: I do agree that like the, and I know some of this is me getting familiar with this work stream.
4442.2 --> 4457.8: But the work on trying to come up with what the taxonomy is for the illicit activity feels exactly like, I'll point to the MITRE ADAPT framework that they presented on, which doesn't get into things like market manipulation on there.
4457.8 --> 4466.1: But it is the first, as far as I know, published, you know, initial taxonomy on especially like crypto specific exploitations.
4466.4 --> 4472.6: So I don't know to what extent we were thinking that this would be like something that would tag on to or continue to expand that.
4472.7 --> 4475.9: Because even though they call it cyber, it's really crypto.
4476.1 --> 4484.2: And the acronym for ADAPT is about like adversarial activity on digital asset payment systems.
4484.3 --> 4486.1: And so this would be like trading systems.
4486.1 --> 4492.0: And so I like, but whatever it is, like it's basically just that mapping and taxonomy.
4492.2 --> 4495.3: So it's a big overlap with that work stream.
4496.7 --> 4505.1: So there's also the challenge I know with AML that there's a lot of activity that is not unique to crypto, like just like definitions of fraud.
4505.8 --> 4524.6: So deciding to what extent does Begin want to kind of focus our efforts on crypto specific, like typology discussions and then the mapping to what that, like if they're specific on chain or like crypto related versions of that, like what does that end up looking like?
4525.3 --> 4549.3: It's a whole issue where like, when I mentioned before that there's no sticks and taxi for AML, it's a, there's no, there's no standards for just illicit finance stuff, generally, not even an agreed upon definition for what different kinds of frauds and scams are, which carry lots of issues for banks because they're interested in the definitions, making sure that they're not held liable in certain, in like authorized push payments and other instances.
4549.5 --> 4551.5: And that's like one of a thousand different considerations.
4551.5 --> 4559.8: So I bring that up to think that we should try to think about what, like what Begin wants to scope and focus the efforts on to be able to define those.
4560.7 --> 4574.2: What we talked about in that first session on information sharing on if we map what the typology is, and then mapping that to like vulnerabilities and controls to defend against it, like a national vulnerability database does in cyber.
4575.1 --> 4588.1: In this case, it's like, I like the idea of machine readable, like basically thinking about what the, like machine readable or

implementation of different like business rules and analytics and heuristics to detect that typology, what that looks like.
4588.2 --> 4594.8: That would be an interesting sort of adaptation in crypto related or on-chain activity pointed to that TTP.
4595.4 --> 4600.6: Like what if you clicked on that in MITRE ADAPT or, or whatever the new framework is, I don't care who's.
4600.6 --> 4606.9: It's also something that would be unique to crypto as well, because of the way that the data structures work.
4607.2 --> 4607.4: Exactly.
4608.1 --> 4613.9: Yeah, I'm noticing a lot of like all the streams of work are converging, which is, which is fun.
4615.2 --> 4615.3: Yeah.
4616.0 --> 4619.8: On the, like, I didn't know, and I know we're at time.
4620.1 --> 4625.0: So to what extent are we, this is just me again, getting familiar with it.
4625.0 --> 4650.8: Are we focusing on the, like, digital forensics, blockchain analytics, like, because you brought up the probabilistic issue and like, there are other interesting precedents of using probabilistic analytics, like in facial recognition and DNA tests and other sorts of things and their use and admissibility or just acceptance generally in like scientific use cases or in approaches around disclosure, those kinds of things.
4650.8 --> 4653.0: Is there, is there an ongoing effort on that?
4653.0 --> 4657.6: Are we kind of more prioritizing the, like, the taxonomy work in this work stream?
4657.8 --> 4673.9: Or is there a lot of stuff that we're focusing on, on like what the playbook would be for digital forensics in crypto for like, I assume a practitioner's guide to like law enforcement on how to think about reliability of different blockchain analytics tools?
4674.1 --> 4676.4: Is that, is that what that work stream is focused on?
4676.9 --> 4680.8: Or do we kind of, do we want to prioritize and focus on the taxonomy one?
4680.9 --> 4682.4: That was a question that I had.
4683.0 --> 4684.1: It's a good question.
4684.3 --> 4693.8: I think we want to try and get as close to the, um, most definable aspect and then expand out.
4694.5 --> 4698.0: Um, so I would say taxonomy would be the focus.
4698.4 --> 4701.9: Happy to be, um, yeah.
4702.8 --> 4711.2: Yeah, I agree with sort of a focus on taxonomy similar to the MITRE framework, but encompassing some of these other.
4712.5 --> 4716.2: Threats as well, beyond just kind of sort of the technical exploits.
4716.4 --> 4722.9: So, um, things like terrorist financing, darknet market usage, those kinds of things as well.
4726.1 --> 4726.6: Yeah.
4726.7 --> 4741.2: Um, just quickly before that, I'm finding in a lot of what the work that I do now, the closer you can get to the most meaningf

ul, least amount of words, the better it is for you to expand out into mu
ltiple different.
4741.2 --> 4743.0: Industries and degrees and stakeholders.
4743.8 --> 4746.6: And I think that's actually echoing in a lot of begin
s work.
4746.9 --> 4749.2: So with the cybersecurity, we're trying to do that.
4749.2 --> 4752.3: We're trying to get like compress these words in this
 work.
4752.4 --> 4753.7: We should also try and do that.
4753.9 --> 4761.2: And then once we get to that point, hopefully we can
be comfortable enough as a group to be like, no, this is the compression
that we can agree on.
4761.2 --> 4770.7: And then each of us individually or as groups can the
n expand out in our own ways and do more expressions of that agreed compr
ession.
4771.7 --> 4772.5: Yeah, that makes sense.
4772.9 --> 4773.6: It does.
4774.0 --> 4775.1: Like, oh no, I'm sorry.
4775.3 --> 4776.1: Sandra's going next.
4776.2 --> 4776.7: No, no, no, no.
4777.5 --> 4777.9: Okay.
4778.1 --> 4778.5: Sorry.
4778.6 --> 4787.5: Since it's directly relevant to that, I, since I know
 that it's called the cybersecurity work stream, but like most of the wor
k that's going on in there isn't at all cyber specific.
4787.5 --> 4791.2: It really is information sharing around crypto activi
ty.
4791.7 --> 4794.9: It includes some cyber related exploits, but a lot of
 it is just broader.
4795.3 --> 4797.6: Like it's illicit financing, crypto related informati
on.
4798.2 --> 4802.7: I feel like, like if we're going to prioritize taxono
my, that it's those two work streams.
4802.9 --> 4803.8: Like it's these combined.
4803.9 --> 4804.8: Yeah, I think you're right.
4806.9 --> 4814.0: So one question I have, and again, it's my lack of kn
owing the background behind some of this work already.
4814.0 --> 4821.0: How much has been mapped across the world of what's a
lready out there versus what new needs to be created?
4821.4 --> 4827.9: Because the challenge that we have found in the last
nine years that we have been in existence, it's not actually a lack of de
finitions.
4828.3 --> 4831.8: It's literally the cacophony of definitions in crypto.
4832.5 --> 4837.5: And I do think the first exercise really does need to
 be the mapping.
4838.0 --> 4841.5: We've done some of that in the last seven years by ju
st collating.

4841.5 --> 4844.5: It's not about GBC creating any of these definitions.
4844.7 --> 4851.6: It's just literally pulling from sources from all ove r the world and then trying to align some of this.
4851.8 --> 4855.2: But it is a huge, huge challenge.
4855.8 --> 4860.0: And I will say to you that I don't know where, how fa r Begin has gone.
4860.1 --> 4871.4: But even now, we sit also on TC307 as Category A and we're looking at the definitions that ISO has put together just to see wh at that looks like mapped to some of the other stuff that we've seen as
4871.5 --> 4872.2: consensus.
4872.6 --> 4876.9: But reality is there is no consensus yet around a lot of these definitions.
4878.4 --> 4880.4: Yeah, well, I think that's the that's the goal.
4880.6 --> 4892.7: So maybe we go with the mapping and then create speci fic targets of compression and then be like this specific, I know scam is the wrong word because it's too broad.
4895.2 --> 4907.9: Rugpull topology has these specific terms and then th en we can expand there and then we can use that as a as a model to overla p with other different typologies.
4909.8 --> 4910.4: Cool.
4910.9 --> 4922.2: Well, it will be in the meeting report and it will be reconstructed and rehypothecated into a few different versions, readable for regulators, business practitioners, these sorts of things.
4922.2 --> 4924.8: So we can kind of start from there.
4927.1 --> 4930.3: About TC307 and vocabulary and the terms defined ther e.
4932.5 --> 4936.9: Don't quote me about ISO TC307, please, outside of th is forum.
4938.4 --> 4948.0: But the way, the way the, yeah I know, but the way th e term has been defined, I mean, although this is about consensus within an ISO committee,
4948.0 --> 4954.4: there is definitely a need to revise to Hammond to ad d.
4955.5 --> 4969.6: And so if we start working on taxonomy, given that BG IN is also a liaison category liaison, we can definitely push for some up dates on this and if there is two liaison organizations doing this and it will be even more impactful.
4971.0 --> 4972.3: Yes, exactly.
4972.3 --> 4983.7: So I have another perspective of why I like the idea to use mathematics instead of natural language, because creating vocabula ry standard takes time.
4985.0 --> 4993.3: So, you know, generally that ISO standard takes minim um two years, but this defining that vocabulary or something like that, i t takes three years, four years.
4993.3 --> 4998.6: So many, many discussions happen in defining terminol ogy or vocabulary.
4999.3 --> 5005.5: But, you know, defining that process or mathematics,

it's much easier for that kind of standards.
5006.4 --> 5014.0: So I, in my experience, I was the project leader and editor of the ISO 29128.
5014.1 --> 5021.6: This is an ISO standard for the formal analysis on the cryptographic protocols.
5022.2 --> 5027.3: So defining the categorization for the mathematics itself, it's very easy to create a standard.
5034.0 --> 5036.4: They are translating mathematics to something.
5041.0 --> 5045.2: Well, AI right now can't understand arithmetic.
5046.1 --> 5051.4: There's certain AI teams out there working on arithmetic and how AI can actually read and understand it.
5051.4 --> 5063.4: But it just associates a set of binary numbers that are close to another set of binary numbers and says one plus one equals two, because that set of binary numbers were associated to that set of binary numbers, which equals three, which is that set of binary numbers.
5063.4 --> 5077.4: I think it actually makes more sense to actually have good English language or any language that's human readable, because that's where it thrives the most and equally can create like a thesaurus based diction out of.
5078.5 --> 5084.3: But I wouldn't say making math for cool math sake, because AI is going to read it someday, maybe in three or four years.
5084.4 --> 5089.0: But right now, I'd say like if you want a high level bath, I wouldn't say right now.
5089.1 --> 5090.5: But I think like taxonomy is great.
5090.6 --> 5091.9: And we've done so well with that so far.
5091.9 --> 5096.1: And I know, Mitchell, you've done amazing things with what you've consumed with your AI.
5096.8 --> 5098.6: Well, I think there are different models.
5098.8 --> 5100.1: Some models are built for math.
5100.6 --> 5102.9: And yeah, they're not very good.
5104.5 --> 5105.7: Some are really good.
5105.9 --> 5111.0: Like some use constrained compute space to solve really hard problems.
5112.1 --> 5113.2: Really cool stuff.
5113.7 --> 5114.7: But what were you going to say, Sun?
5114.7 --> 5122.2: Actually, you guys have said flavors of what I was going to say, which is we probably need all of those things.
5122.3 --> 5132.7: Because if we're going to be helpful to regulators in a given jurisdiction, we also have to align the definitions to how they're seeing the legislation and or rules that are being implemented.
5132.8 --> 5137.0: And sometimes they're not going to be able to be hard coded in a concise mathematical way.
5137.1 --> 5138.7: And that will be the challenges.
5138.9 --> 5140.6: We probably need a combination of all the above.
5140.6 --> 5156.1: Yeah, I think perhaps the point that like the lowest hanging fruit could potentially be trying to find one of these taxonomies that equal math and then show that that is possible.

5156.7 --> 5156.9: Sure.
5157.1 --> 5168.0: And then expand out from there and be like, these are
 the deviations of this because of complexity and because humans are addi
ng randomness to this.
5168.0 --> 5171.2: But, yeah, I like it as a starting point.
5171.6 --> 5177.6: I think it's something that we can just do and then s
end it.
5187.4 --> 5190.5: I definitely think, like, I see the value.
5190.7 --> 5194.6: Like, there's a reason why FinCEN published advisorie
s with red flags in it all the time.
5194.6 --> 5198.7: We wanted to help inform industry about what we're se
eing and what's helpful.
5199.8 --> 5210.0: I don't know to what extent that any of the analytics
 firms will feel that flagging what the digital signals are to detect tha
t stuff is them forking over proprietary information and other things.
5210.1 --> 5214.7: So anyway, but that's going to be a reality for any i
ssue that comes up.
5214.8 --> 5216.3: And so, like, I shouldn't deter us.
5216.3 --> 5218.7: It's just I recognize, like, anyway.
5218.7 --> 5234.5: But either way, I think that that translation from di
scussing what the different typologies are to, like, including the, like,
 as much as we can, a machine readable way to discuss what the vulnerabil
ity is that you're looking for, what it looks like when it's being exploi
ted.
5236.1 --> 5251.9: And then, like, what the technical fixes, which creat
es a really interesting, unique version of an NVD, of a National Vulnerab
ility Database and stuff, but for crypto and blockchain in a really valua
ble way.
5252.1 --> 5253.5: So I think that's good.
5254.3 --> 5261.9: If there is work, whenever there is work on the, like,
 practitioner's guide for using analytics, I think, I don't know if we've
 been using Daubert and other things.
5261.9 --> 5270.9: Like, I have a lot of thoughts about ways to structur
e based on other precedents for, like, use and admissibility of expert te
stimony and digital evidence.
5271.1 --> 5272.7: Although I'm sure you've been working with Amanda.
5272.8 --> 5274.8: I'm sure all that has probably been accounted for alr
eady.
5274.9 --> 5276.4: So, OK, never mind then.
5276.7 --> 5277.1: Thanks.
5286.9 --> 5292.7: And so, as I mentioned, so there is a network recepti
on at 5.30pm.
5293.5 --> 5296.4: So the venue is just five minutes walk from here.
5297.3 --> 5302.0: And we will bring, so we will depart here at 5.20ish.
5302.9 --> 5306.0: And so please take 10 minutes to get here.
5306.0 --> 5307.8: But anyway, thank you very much for coming today.